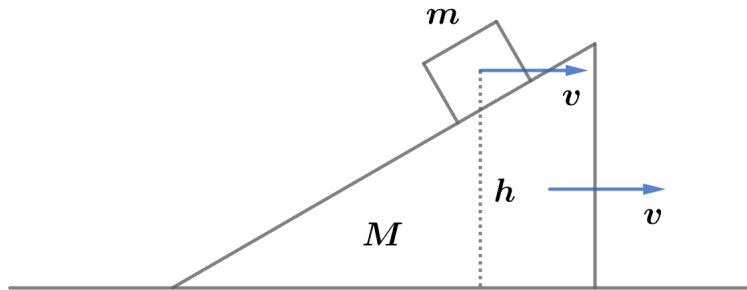


2010 F=ma Exam: Problem 15

Kevin S. Huang



When the block reaches its maximum height, it is at rest with respect to the incline. Thus, they both have the same velocity v . Conserving linear momentum,

$$mv_0 = (m + M)v$$
$$v = \frac{mv_0}{m + M}$$

To find the height h , we conserve energy,

$$\frac{1}{2}mv_0^2 = mgh + \frac{1}{2}(m + M)v^2 = mgh + \frac{m^2v_0^2}{2(m + M)}$$
$$h = \frac{v_0^2}{2g} \left(1 - \frac{m}{m + M} \right) = \frac{Mv_0^2}{2g(m + M)}$$

so the answer is C.